validate_taxa_classification_coverage.R

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`validate_taxa_classification_coverage` *Validate taxonomic-classification coverage*

Description

Validates that all community taxa have been classified. When missing taxa are found, stops with an informative error message listing the count of unclassified taxa. Returns 'TRUE' invisibly when no taxa are missing.

Usage

```
validate_taxa_classification_coverage(
  vec_taxa_without_classification,
  file_missing_taxa_template = NULL
)
```

Arguments

`vec_taxa_without_classification`
: A character vector of taxon names that could not be classified automatically or via the auxiliary table. An empty vector ('character(0)') signals full coverage and causes the function to return silently.

`file_missing_taxa_template`
: 'NULL' by default. Optional placeholder for a targets file dependency. When supplied in a pipeline, it ensures the missing taxa CSV is written before the error is raised.

Details

The missing taxa are stored as a targets object ‘data_missing_taxa_template‘ in the pipeline store. Inspect them with ‘targets::tar_read("data_missing_taxa_template")‘. During a pipeline run the same rows are also appended to ‘Data/Input/missing_taxa_template.csv‘ for manual review. The object is a ‘tibble‘ with columns ‘sel_name‘, ‘kingdom‘, ‘phylum‘, ‘class‘, ‘order‘, ‘family‘, ‘genus‘, and ‘species‘; rank columns are left as ‘NA‘ for manual completion. Fill in the missing classifications and copy or append rows to ‘Data/Input/aux_classification_table.csv‘, then re-run the pipeline. Use the helper script in ‘R/03_Supplementary_analyses/Classification/‘ to coalesce templates across all pipeline stores into one CSV.

Value

‘TRUE‘ invisibly when every taxon is classified. Stops with an error when any taxa are missing.

See Also

[load_auxiliary_classification_table()], [build_combined_classification_table()], [select_unclassified_taxa()]

Examples

```
validate_taxa_classification_coverage(
  vec_taxa_without_classification = base::character(0)
)

base::try(
  validate_taxa_classification_coverage(
    vec_taxa_without_classification = base::c("Taxon_a")
  )
)
```

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